Cmod A7-35T Pin Specification

1. DIP Connector Overview

The Cmod A7-35T has a 48-pin DIP connector (J1). All digital I/O pins operate at **LVCMOS33** (3.3 V logic level) with **no series resistors**.

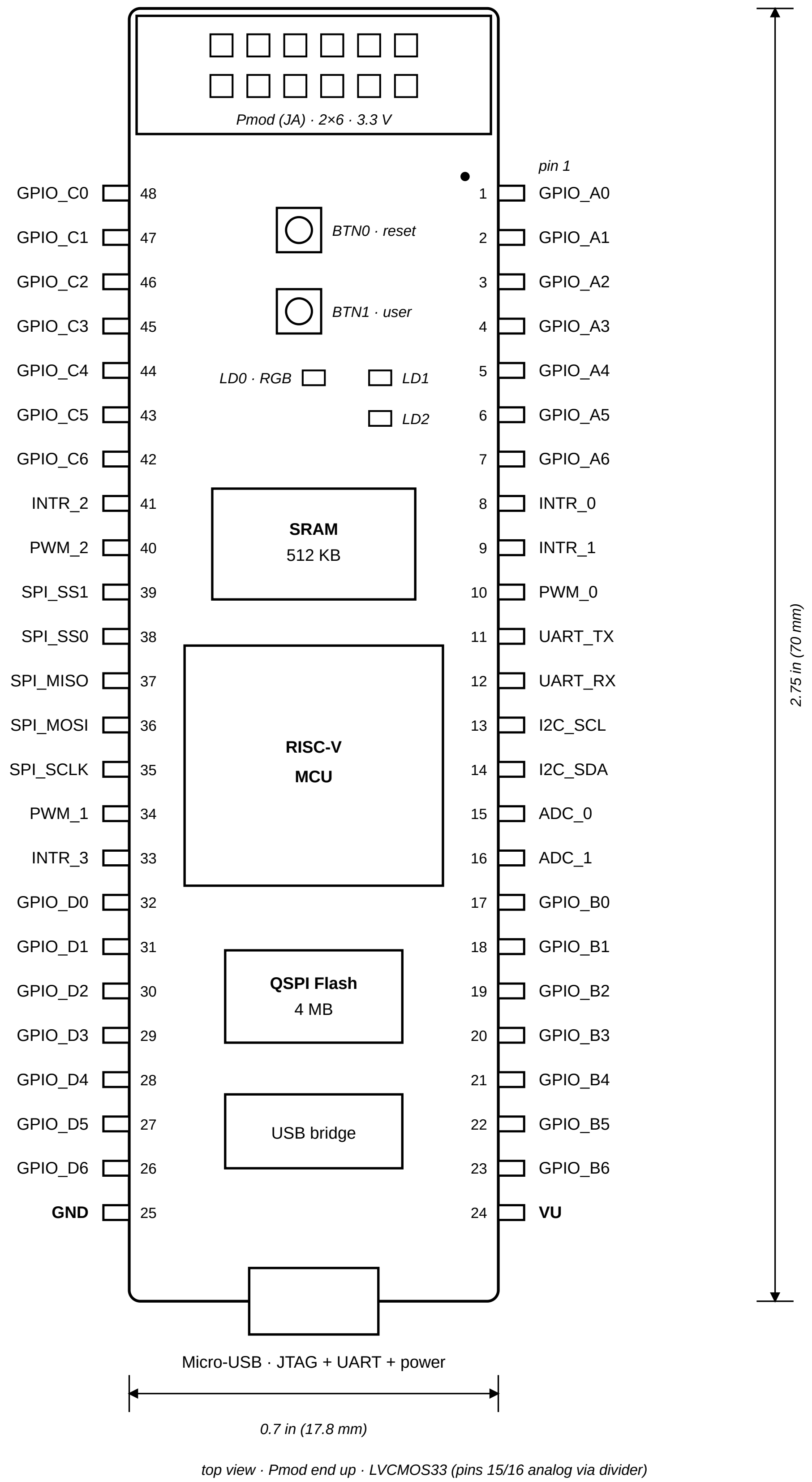

Figure 1. Cmod A7-35T DIP pinout (top view)

Category	Count
GPIO (4 groups × 7-bit)	28
External Interrupts	4

Category	Count
PWM Outputs	3
UART 1 (TX + RX)	2
I2C (SCL + SDA)	2
SPI (SCLK/MOSI/MISO/SS×2)	5
Analog Inputs (XADC)	2
Power (VU + GND)	2
Total	**48**

Peripheral registers and base addresses are documented in the IP peripheral reference.

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2. Pin Map — Left Side (Pin 1–24)

DIP Pin	Signal Name	FPGA Pin	Direction	Function
1	GPIO_A0	M3	I/O	General purpose GPIO, Group A bit 0
2	GPIO_A1	L3	I/O	General purpose GPIO, Group A bit 1
3	GPIO_A2	A16	I/O	General purpose GPIO, Group A bit 2
4	GPIO_A3	K3	I/O	General purpose GPIO, Group A bit 3
5	GPIO_A4	C15	I/O	General purpose GPIO, Group A bit 4
6	GPIO_A5	H1	I/O	General purpose GPIO, Group A bit 5
7	GPIO_A6	A15	I/O	General purpose GPIO, Group A bit 6
8	INTR_0	B15	Input	External interrupt input 0
9	INTR_1	A14	Input	External interrupt input 1
10	PWM_0	J3	Output	PWM output channel 0
11	UART_TX	J1	Output	UART 1 transmit
12	UART_RX	K2	Input	UART 1 receive
13	I2C_SCL	L1	I/O (open-drain)	I2C clock (external 4.7 kΩ pull-up recommended)
14	I2C_SDA	L2	I/O (open-drain)	I2C data (external 4.7 kΩ pull-up recommended)
15	ADC_0	G3 / G2	Analog	XADC VAUX4 (ain_p[15] / ain_n[15])
16	ADC_1	H2 / J2	Analog	XADC VAUX12 (ain_p[16] / ain_n[16])

DIP Pin	Signal Name	FPGA Pin	Direction	Function
17	GPIO_B0	M1	I/O	General purpose GPIO, Group B bit 0
18	GPIO_B1	N3	I/O	General purpose GPIO, Group B bit 1
19	GPIO_B2	P3	I/O	General purpose GPIO, Group B bit 2
20	GPIO_B3	M2	I/O	General purpose GPIO, Group B bit 3
21	GPIO_B4	N1	I/O	General purpose GPIO, Group B bit 4
22	GPIO_B5	N2	I/O	General purpose GPIO, Group B bit 5
23	GPIO_B6	P1	I/O	General purpose GPIO, Group B bit 6
24	**VU**	—	Power	Power input (ext) / Power output (USB)

3. Pin Map — Right Side (Pin 25–48)

DIP Pin	Signal Name	FPGA Pin	Direction	Function
25	**GND**	—	Power	Ground reference
26	GPIO_D6	R3	I/O	General purpose GPIO, Group D bit 6
27	GPIO_D5	T3	I/O	General purpose GPIO, Group D bit 5
28	GPIO_D4	R2	I/O	General purpose GPIO, Group D bit 4
29	GPIO_D3	T1	I/O	General purpose GPIO, Group D bit 3
30	GPIO_D2	T2	I/O	General purpose GPIO, Group D bit 2
31	GPIO_D1	U1	I/O	General purpose GPIO, Group D bit 1
32	GPIO_D0	W2	I/O	General purpose GPIO, Group D bit 0
33	INTR_3	V2	Input	External interrupt input 3
34	PWM_1	W3	Output	PWM output channel 1
35	SPI_SCLK	V3	Output	SPI clock (software-settable, up to 25 MHz)
36	SPI_MOSI	W5	Output	SPI master-out slave-in
37	SPI_MISO	V4	Input	SPI master-in slave-out
38	SPI_SS0	U4	Output	SPI slave select 0 (active low)
39	SPI_SS1	V5	Output	SPI slave select 1 (active low)
40	PWM_2	W4	Output	PWM output channel 2
41	INTR_2	U5	Input	External interrupt input 2
42	GPIO_C6	U2	I/O	General purpose GPIO, Group C bit 6

DIP Pin	Signal Name	FPGA Pin	Direction	Function
43	GPIO_C5	W6	I/O	General purpose GPIO, Group C bit 5
44	GPIO_C4	U3	I/O	General purpose GPIO, Group C bit 4
45	GPIO_C3	U7	I/O	General purpose GPIO, Group C bit 3
46	GPIO_C2	W7	I/O	General purpose GPIO, Group C bit 2
47	GPIO_C1	U8	I/O	General purpose GPIO, Group C bit 1
48	GPIO_C0	V8	I/O	General purpose GPIO, Group C bit 0

4. GPIO Groups

Group	Width	DIP Pins
A	7-bit	1–7
B	7-bit	17–23
C	7-bit	42–48
D	7-bit	26–32

5. Interrupt Inputs

Signal	DIP Pin	FPGA Pin
INTR_0	8	B15
INTR_1	9	A14
INTR_2	41	U5
INTR_3	33	V2

6. PWM Outputs

Signal	DIP Pin	FPGA Pin
PWM_0	10	J3
PWM_1	34	W3
PWM_2	40	W4

7. UART Interfaces

Interface	TX Pin	RX Pin	TX FPGA	RX FPGA	Connection

Interface	TX Pin	RX Pin	TX FPGA	RX FPGA	Connection
UART 0 (USB)	—	—	J18	J17	Via Micro-USB (J3), no DIP pin
UART 1 (External)	DIP 11	DIP 12	J1	K2	Exposed on DIP connector

8. Analog Inputs (XADC)

Signal	DIP Pin	FPGA Pin (P/N)	XADC Channel
ADC_0	15	G3 / G2	VAUX4
ADC_1	16	H2 / J2	VAUX12

Analog Input Circuit

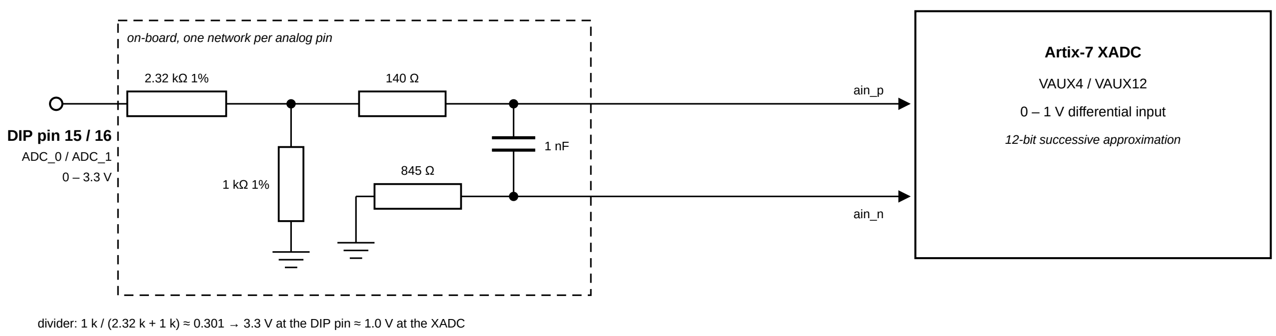

Figure 2. On-board voltage divider circuit for XADC analog inputs

The XADC expects an input range of 0–1 V. The board includes a resistive voltage divider (2.32 KΩ / 1 KΩ, all 1% precision) that scales the DIP pin voltage down to the FPGA's acceptable range. A 140 Ω series resistor and 1 nF capacitor are placed on the differential pair for filtering. The ADx_N path has an 845 Ω series resistor.

Parameter	Value
Max input voltage on DIP pin 15/16	**3.3 V** (relative to GND on pin 25)
Voltage at FPGA after divider	0–1 V
Divider ratio	1 KΩ / (2.32 KΩ + 1 KΩ) ≈ 0.301
Resistor tolerance	1%

> **Note:** Pins 15 and 16 are routed through on-board voltage dividers. If used as analog inputs, they must **not** be assigned as digital I/O in the constraints file. Do not exceed 3.3 V on these pins.

9. Serial Expansion Pins (I2C / SPI)

DIP Pin	FPGA Pin	Signal	Description
13	L1	I2C_SCL	I2C clock, 100 kHz
14	L2	I2C_SDA	I2C data
35	V3	SPI_SCLK	SPI clock (software-settable, up to 25 MHz)
36	W5	SPI_MOSI	Master out, slave in
37	V4	SPI_MISO	Master in, slave out
38	U4	SPI_SS0	Slave select 0 (active low)
39	V5	SPI_SS1	Slave select 1 (active low)

> **I2C pull-ups:** SCL/SDA are open-drain. Weak FPGA-internal pull-ups are enabled as a fallback, but connect external 4.7 kΩ resistors to 3.3 V when wiring real devices.

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10. On-Board I/O (No DIP Pin Exposure)

Function	FPGA Pins
LEDs (2)	A17, C16
User button (BTN1)	B18
Reset button (BTN0)	A18
RGB LED	B17, B16, C17

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11. Electrical Characteristics

Parameter	Value
I/O Standard	LVCMOS33 (3.3 V)
Series Resistance on DIP Pins	None
Max Input Voltage (digital I/O)	abs. max –0.4 V to VCCO + 0.55 V = **3.85 V** (DS181); not 5 V tolerant
Analog Input Range (Pin 15, 16)	0–3.3 V (on-board divider scales to the XADC's 0–1 V)
Clock Source	12 MHz oscillator (FPGA pin L17), PLL → 100 MHz